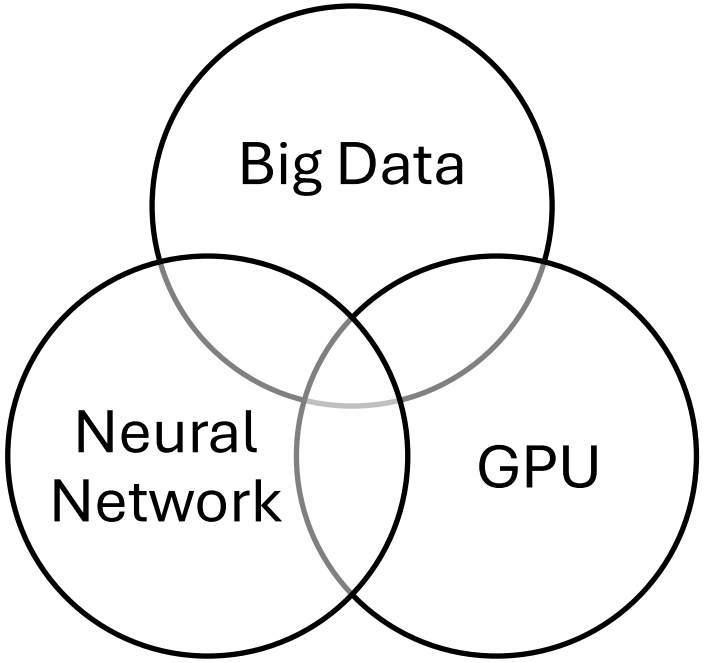
Introduction to Mutimodal AI for Biomedical Applications

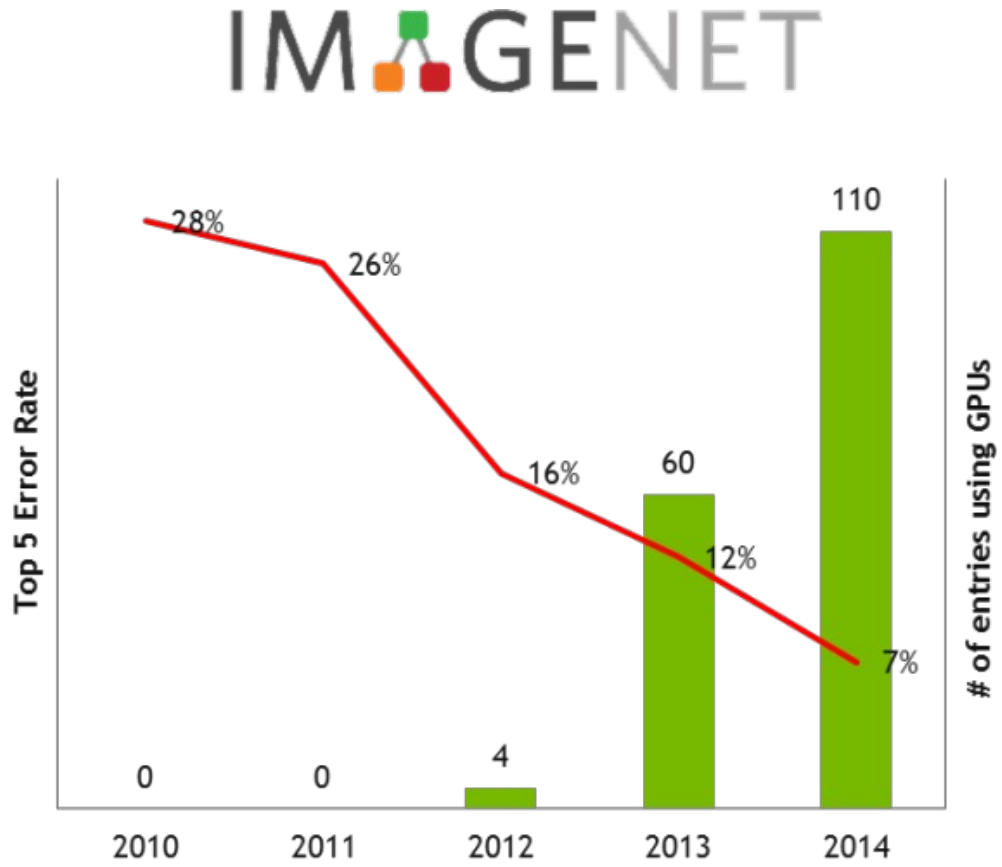
[W2-1: Introduction to Neural Network]

Jun-Hyuk Kim

Department of AI

Deep Learning Revolution Begins



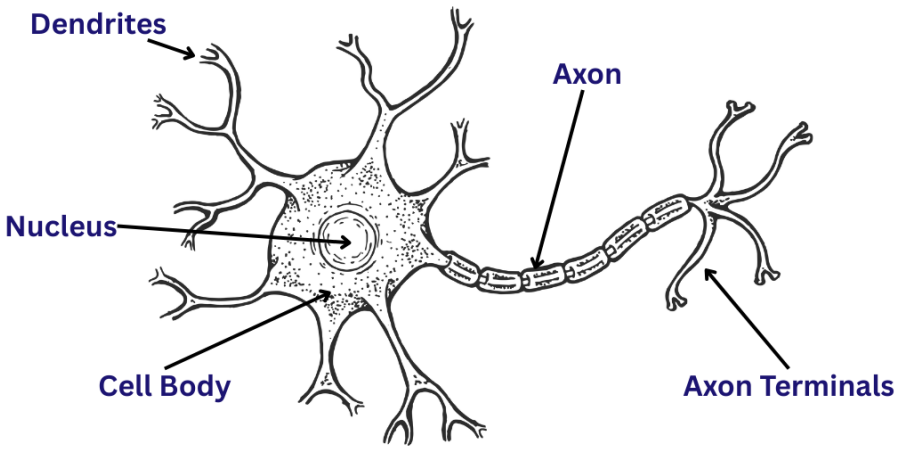


Overview

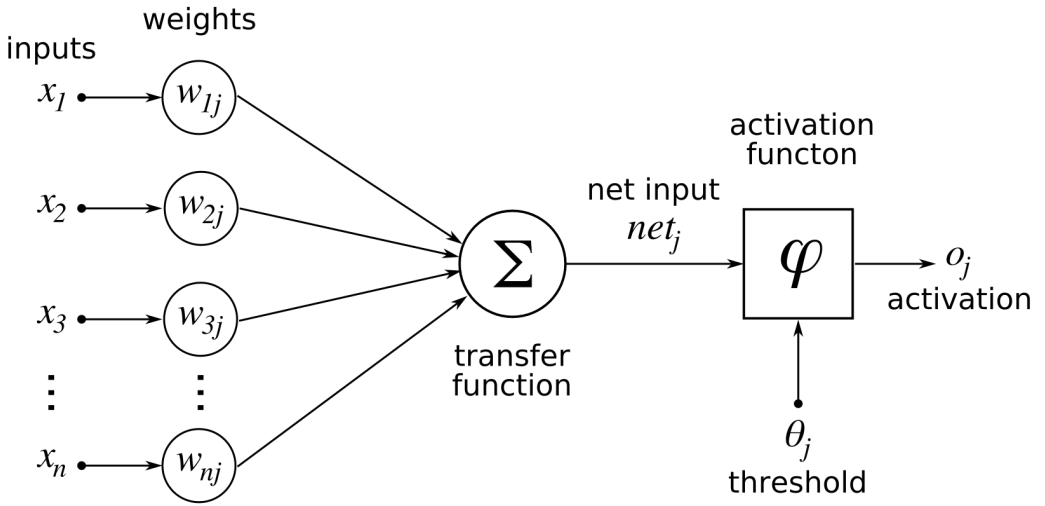
- From Biological to Artificial Neurons
- From Shallow to Deep Neural Networks
  - Architecture
  - Training
  - Computation
  - Data

From Biological to Artificial Neurons

Biological Neuron



Artificial Neuron



1873 – Seeing the Neuron

- Camillo Golgi developed Golgi staining
  - Called “black reaction”
  - Made individual nerve cells visible in remarkable detail
- Golgi’s view – Reticular theory
  - The nervous system forms a continuous interconnected network





1888 – Neurons as Individual Units

- Santiago Ramón y Cajal
  - Used Golgi's staining technique to study the nervous system
- Neuron Doctrine
  - The nervous system is composed of separate, individual cells called neurons





Two Views of the Nervous System



Camillo Golgi

Reticular Theory

VS.



Santiago Ramón y Cajal

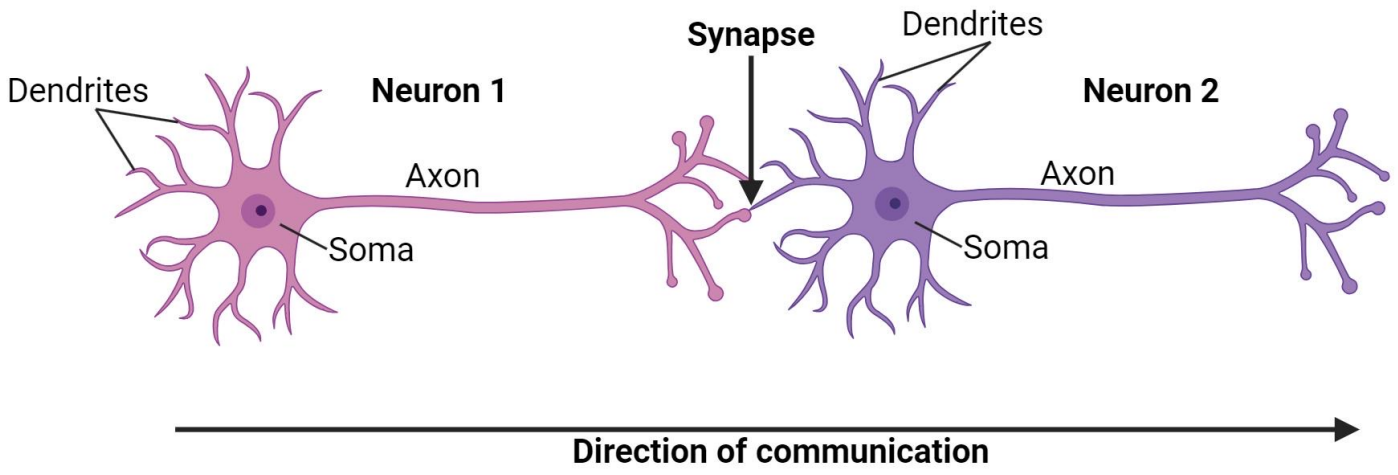
Neuron Doctrine

In 1906, both shared the Nobel Prize in Physiology or Medicine

→ Today, the neuron doctrine is the foundation of modern neuroscience.

1897 – Connecting the Neurons

- Charles Sherrington introduced the term “synapse”
- Neurons communicate through specialized junctions called synapses

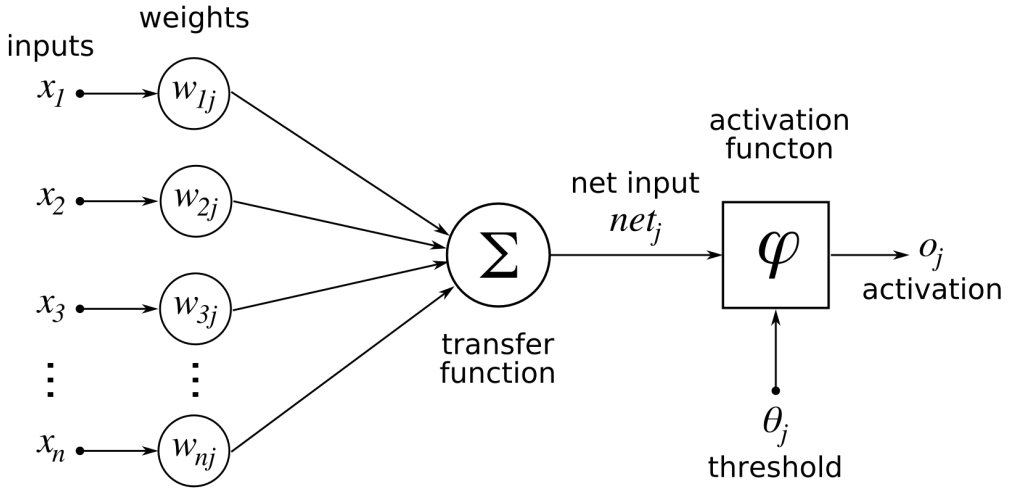




1943 – Modeling the Neurons

- Warren McCulloch & Walter Pitts
  - Proposed the mathmatical model of an artificial neuron (M-P neuron)
  - Binary inputs (0 or 1)
  - Fixed connections (weights)
  - Threshold-based activation
  - Binary outputs (0 or 1)





1943 – Modeling the Neurons

Weights Threshold

$1 \times x_1 + 1 \times x_2 \geq 2 \rightarrow y = 1$

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

→ AND

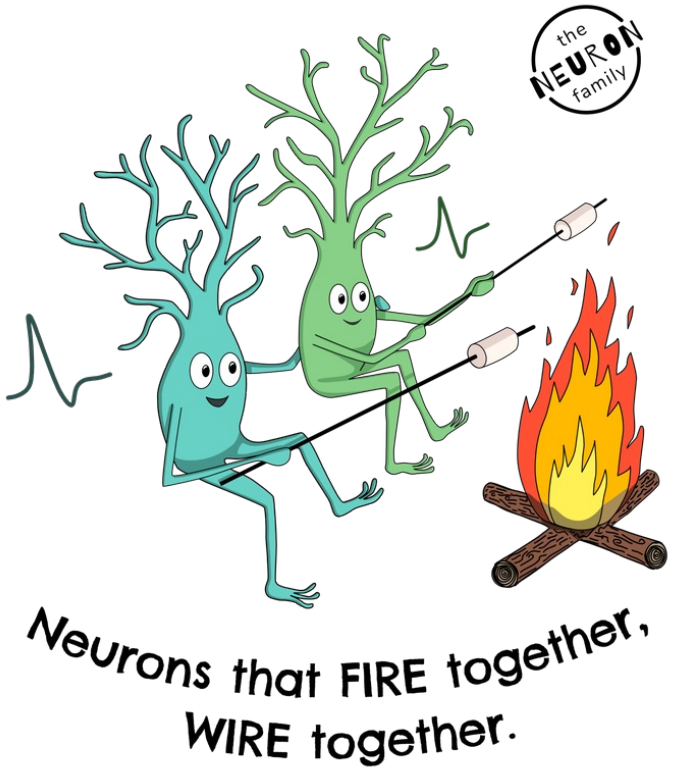
$1 \times x_1 + 1 \times x_2 \geq 1 \rightarrow y = 1$

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

→ OR

1949 – Learning through Connections

- Donald Hebb
  - Proposed a biological theory of learning
  - Learning occurs through changes in the connections between neurons
  - Repeated activation can strengthen neural connections



1958 – Learning from Data

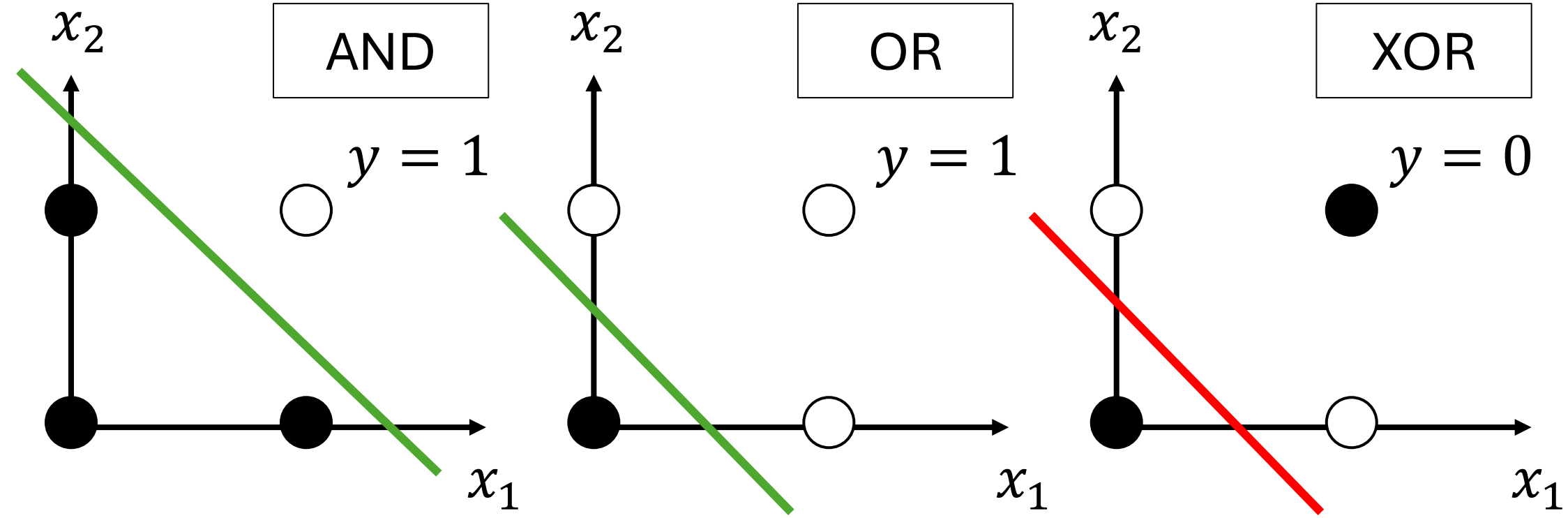
- Frank Rosenblatt
  - Introduced the Perceptron, which has a similar basic structure to the M-P neuron but uses learnable weights
  - Adjusted weights based on prediction errors
  - Enabled a machine to learn from data



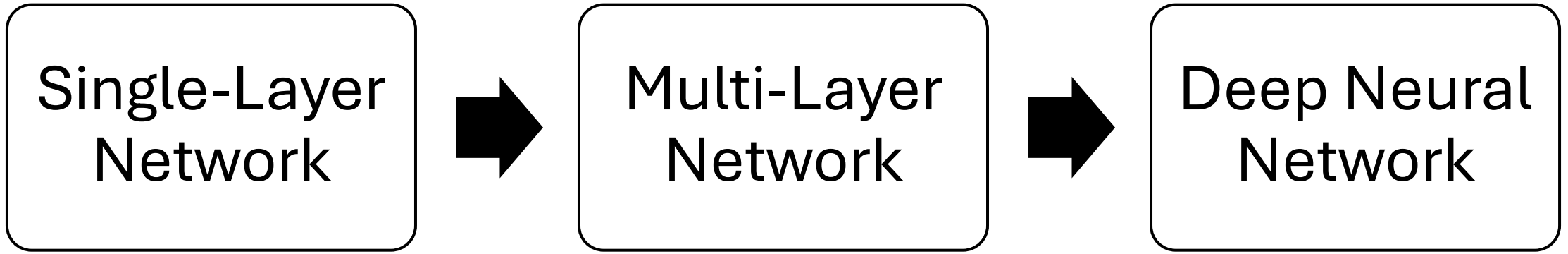
1969 – The Limits of the Perceptron

Linearly separable

Not linearly separable



From Shallow to Deep Neural Networks



What made deep neural networks possible?

Beyond a Single Layer

Single-Layer Perceptron

- Input → Output
- XOR ❌

Multi-Layer Perceptron

- Input → Hidden Layer → Output
- XOR ✅

Hidden layers allow neural networks

to represent more <u>complex, non-linear functions</u>

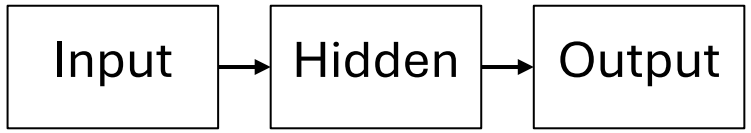
→ But how can we train the hidden layers?

1970 – Propagating Information Backward

- Seppo Linnainmaa
  - Developed reverse-mode automatic differentiation
  - Provided a way to trace how earlier computation affect the output
  - Became a mathematical foundation of modern backpropagation

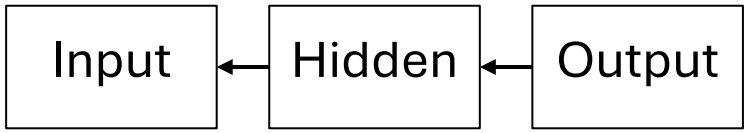
Forward

Make a predcition



Backward

Determine how internal connections should change

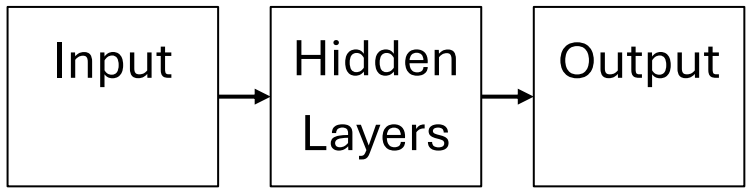


1974 – Learning from Errors

- Paul Werbos
  - Applied backward differentiation to neural network learning
  - Enabled errors to guide learning in earlier layers

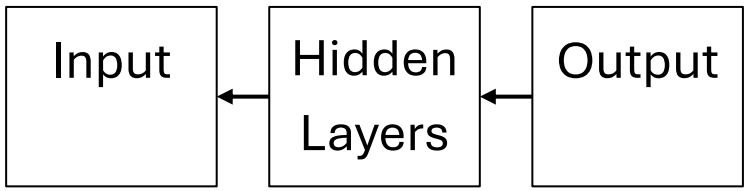
Forward

Make a predcition



Backward

Determine how hidden layers should change



1986 – Learning through Multiple Layers

- David Rumelhart, Geoffrey Hinton & Ronald Williams
  - Popularized backpropagation for training multi-layer networks
  - Showed that hidden layers can learn useful representations



Letter | Published: 09 October 1986

Learning representations by back-propagating errors

David E. Rumelhart, Geoffrey E. Hinton & Ronald J. Williams

Nature **323**, 533–536 (1986) | Cite this article



Are Neural Networks Solved?

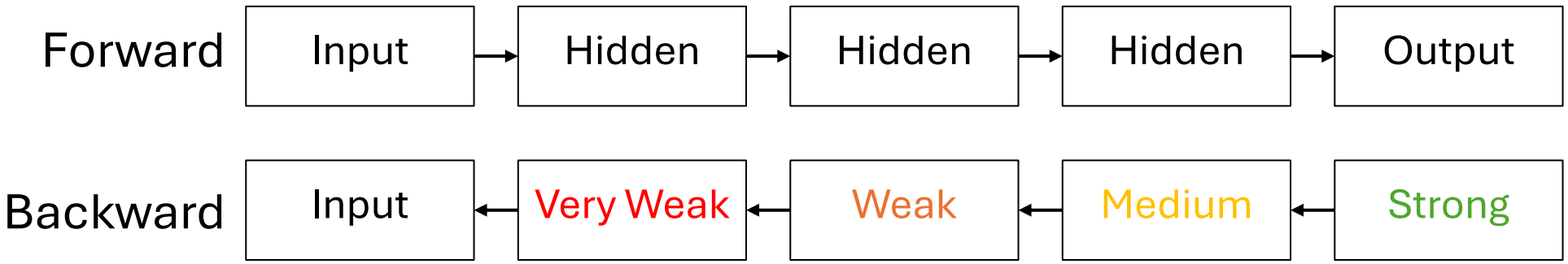
- More layers → More powerful representations
- Backpropagation → Train multiple layers
- Then, can we just go deeper?



Not that easy… → Vanishing gradient problem

1991 – The Vanishing Gradient Problem

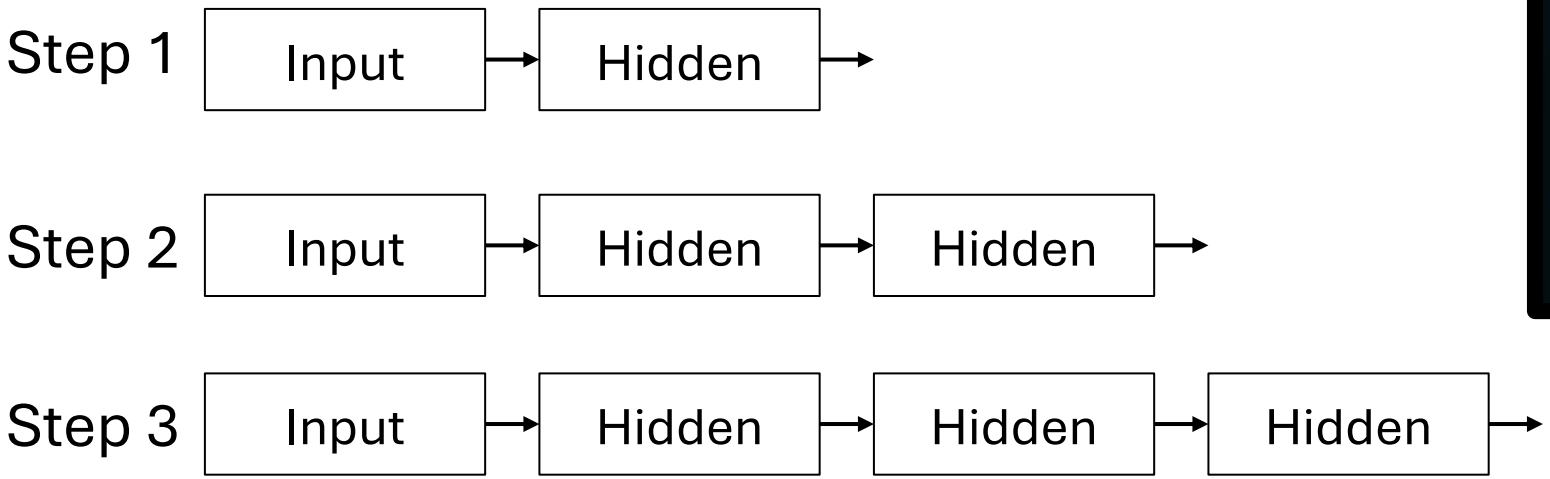
- Sepp Hochreiter
  - Training information must travel backward through many layers
  - This learning signal can become smaller and smaller
  - Earlier layers may receive almost no useful learning signal



2006 – The Revival of Deep Learning

- Geoffrey Hinton et al.
  - Introduced an effective way to train deep neural networks
  - Trained the network one layer at a time



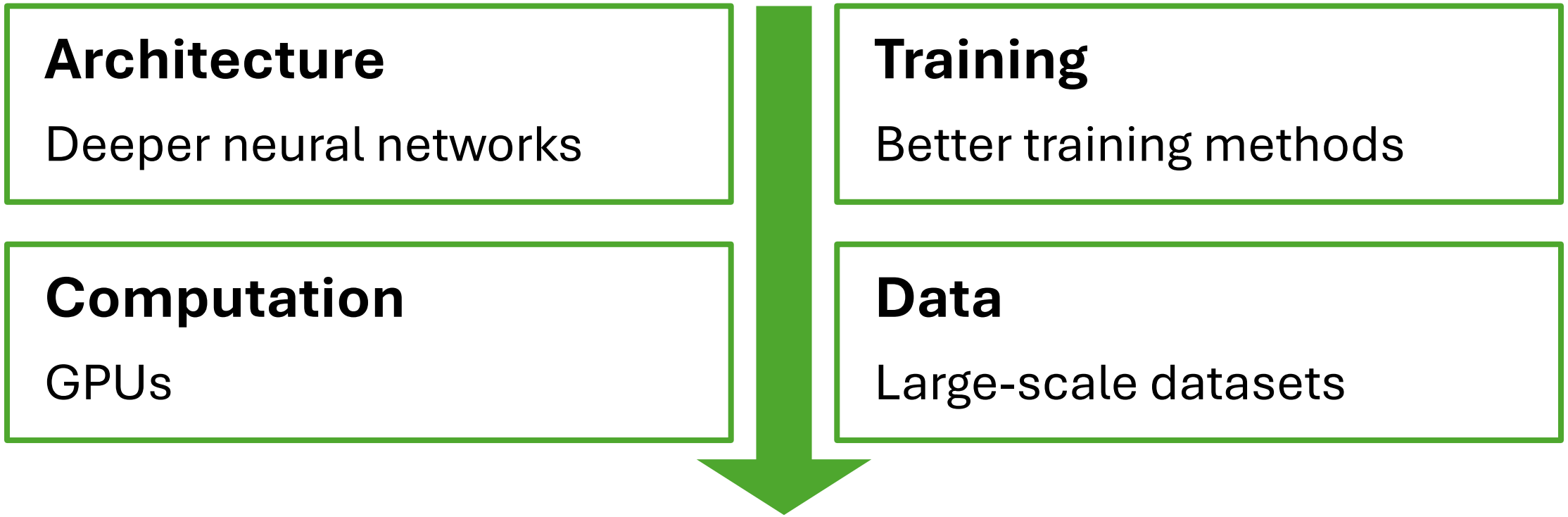


Late 2000s – Better Training for Deep Networks

- Better initialization → Better starting points
- Better activation functions → Improved learning through many layers
- Better training techniques → More effective training

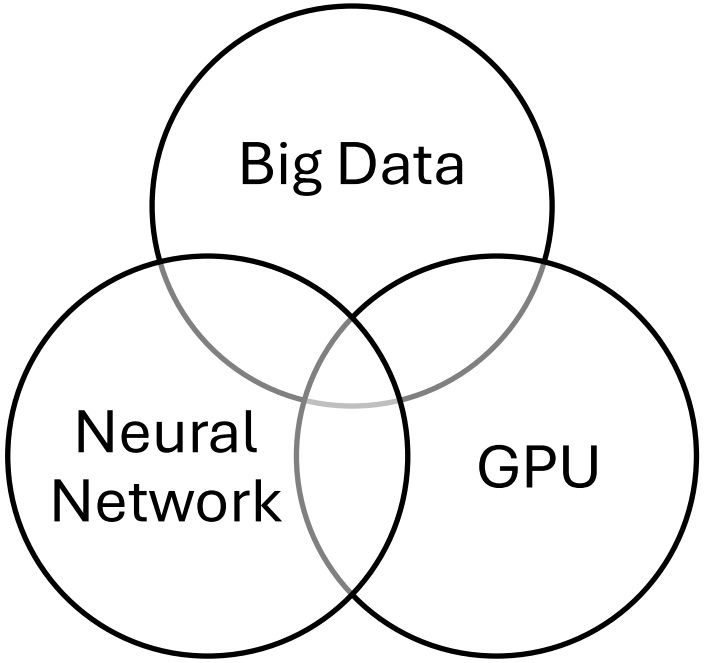
→ Deep networks became easier to train.

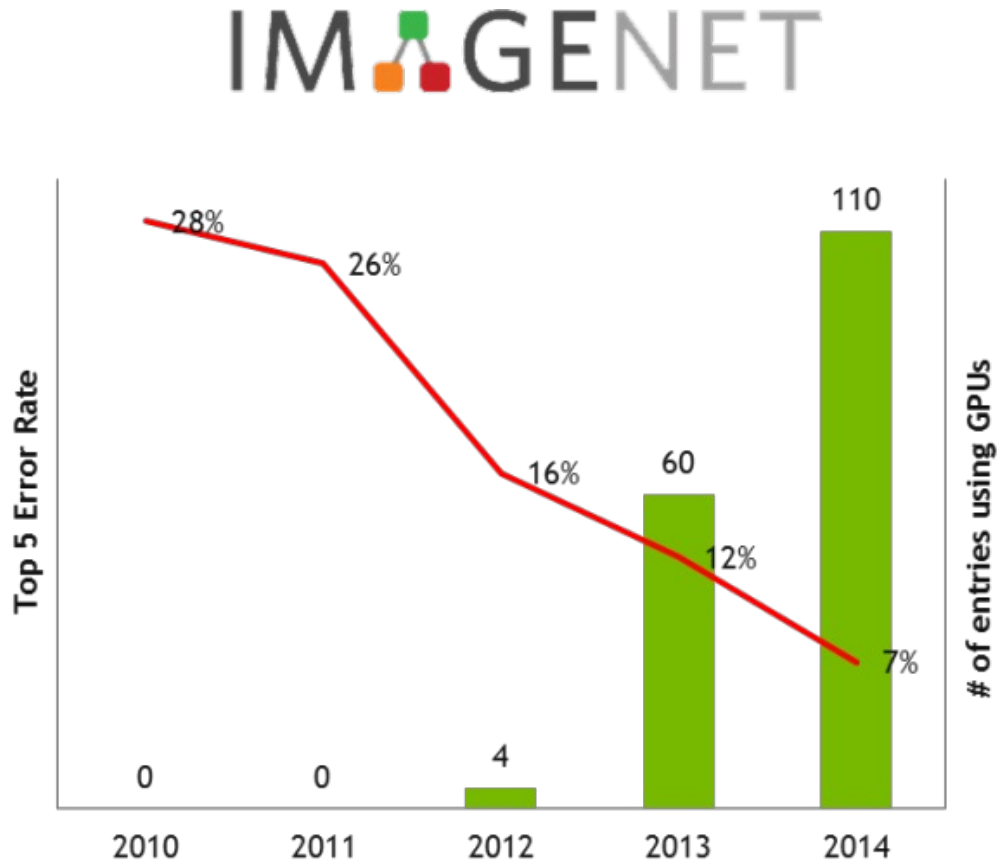
The Pieces Were Coming Together



Read for the Deep Learning Revolution?

2012 – Deep Learning Revolution Begins





The Journey to Deep Neural Networks

1873

Golgi

Neuron visualization

1888

Cajal

Neuron Doctrine

1897

Sherrington

Synapse

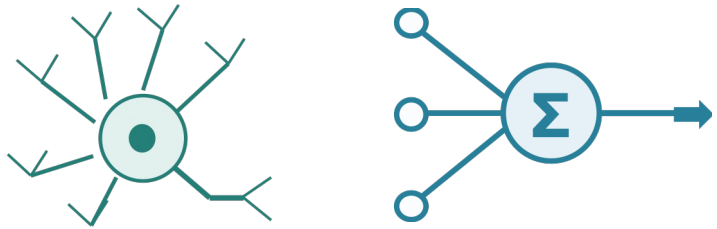
1949

Hebb

Learning through connections

Biological Inspiration

From biological to artificial neurons



1943

McCulloch & Pitts

Artificial neuron

1958

Rosenblatt

Perceptron

The Journey to Deep Neural Networks

GPUs, Big Data

1958
Rosenblatt
Perceptron

1886
Hinton et al .
Backpropagation

2006
Hinton et al.
Deep Belief Network

2012
Hinton et al.
ImageNet Challenge Winner

Challenges

1969
Minskey & Papert
XOR Problem

1991
Hochreiter
Vanishing Gradient Problem

From Shallow to Deep Networks

